

Statistical Modeling

DATA.STAT.740

Lecture 4

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Generalized linear models

$\mathbf{y}_i \sim p(y \mid X)$ Random component

$\mu_i = x_i^T \beta$ Systematic component

$\mu_i = g(\mathbb{E}(\mathbf{y}_i))$ Link function

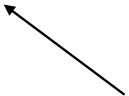
Generalized linear models

Let's change the notation a bit (reserve $\mu_i = E(\mathbf{y}_i)$)

$\mathbf{y}_i \sim p(y \mid X)$ Random component

$\eta_i = x_i^T \beta$ Systematic component

$\eta_i = g(\mu_i)$ Link function


$$\mu_i = E(\mathbf{y}_i)$$

Logistic regression

Observations are Bernoulli-distributed

$$\mathbf{y}_i \sim \text{Ber}(\mu_i) = f(y_i \mid \mu_i) = \mu^{y_i} (1 - \mu_i)^{1-y_i}$$

$$\text{E}(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \mu_i(1 - \mu_i)$$

Bernoulli distribution is suitable for modelling binary outcome (success/failure)

Most often used link function is the logit link

Example: Logistic-regression

Let's simulate data from logistic regression model

$$\mathbf{y}_i \sim \text{Ber}(\mu_i) = f(y_i \mid \mu_i) = \mu^{y_i} (1 - \mu_i)^{1-y_i}$$

$$\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$

$$\mu_i = g^{-1}(\eta_i) = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

$$\mathbb{E}(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \mu_i(1 - \mu_i)$$

Poisson regression

Observations are Poisson-distributed

$$\mathbf{y}_i \sim \text{Poisson}(\mu_i) = f(y_i \mid \mu_i) = \frac{\mu_i^{y_i} \exp(-\mu_i)}{y_i!}$$
$$\text{E}(\mathbf{y}_i) = \mu_i$$
$$\text{Var}(\mathbf{y}_i) = \mu_i$$

Poisson distribution is suitable for count outcome (0, 1, 2, ...)

Most often used link function is the log link

Example: Poisson-regression

Let's simulate data from Poisson regression model

$$\mathbf{y}_i \sim \text{Poisson}(\mu_i) = f(y_i \mid \mu_i) = \frac{\mu_i^{y_i} \exp(-\mu_i)}{y_i!}$$

$$\eta_i = x_i^T \beta$$

$$\mu_i = g^{-1}(\eta_i) = \exp(\eta_i)$$

$$\mathbb{E}(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \mu_i$$

Gamma regression

Observations are Gamma-distributed ($y_i > 0, \phi > 0$)

$$y_i \sim \text{Gamma}(\mu_i, \phi) = f(y_i | \mu_i, \phi) = \frac{1}{\Gamma(\frac{1}{\phi})} \left(\frac{1}{\phi \mu_i} \right)^{\frac{1}{\phi}} y_i^{\frac{1}{\phi}-1} \exp \left(-\frac{y_i}{\phi \mu_i} \right)$$

$$E(y_i) = \mu_i$$

$$\text{Var}(y_i) = \phi \mu_i^2$$

Gamma distribution is suitable in modeling situations when outcome is positive and variance increases proportionally to μ_i^2 (heteroscedastic noise)

Most often used link function is the log link (also inverse link)

Example: Gamma-regression

Let's simulate data from Gamma regression model

$$\mathbf{y}_i \sim \text{Gamma}(\mu_i, \phi) = f(y_i \mid \mu_i, \phi) = \frac{1}{\Gamma(\frac{1}{\phi})} \left(\frac{1}{\phi \mu_i} \right)^{\frac{1}{\phi}} y_i^{\frac{1}{\phi}-1} \exp \left(-\frac{y_i}{\phi \mu_i} \right)$$

$$\eta_i = x_i^T \beta$$

$$\mu_i = g^{-1}(\eta_i) = \exp(\eta_i)$$

$$\mathbb{E}(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \phi \mu_i^2$$

Example: Gamma-regression

Let's simulate data from Gamma regression model

$$y_i \sim \text{Gamma}(\mu_i, \phi) = f(y_i \mid \mu_i, \phi) = \frac{1}{\Gamma(\frac{1}{\phi})} \left(\frac{1}{\phi \mu_i} \right)^{\frac{1}{\phi}} y_i^{\frac{1}{\phi}-1} \exp \left(-\frac{y_i}{\phi \mu_i} \right)$$

$$\eta_i = x_i^T \beta$$

$$\mu_i = g^{-1}(\eta_i) = \exp(\eta_i)$$

$$E(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \phi \mu_i^2$$

Compare the common gamma distribution formulation (used in R) and the mean & dispersion - formulation above

Common:

$$f(y_i \mid \alpha, \theta_i) = \frac{1}{\Gamma(\alpha) \theta_i^\alpha} y_i^{\alpha-1} \exp \left(-\frac{y_i}{\theta_i} \right)$$

where

$$\alpha = \frac{1}{\phi}$$

$$\theta_i = \mu_i \phi$$

Beta regression (not really a GLM but close enough)

Observations are Beta-distributed $(0 < y_i < 1, \phi > 0)$

$$y_i \sim \text{Beta}(\mu_i, \phi) = f(y_i \mid \mu_i, \phi) = \frac{\Gamma(\phi)}{\Gamma(\mu_i\phi)\Gamma((1-\mu_i)\phi)} y_i^{\mu_i\phi-1} (1-y_i)^{(1-\mu_i)\phi-1}$$

$$E(y_i) = \mu_i$$

$$\text{Var}(y_i) = \frac{\mu_i(1-\mu_i)}{1+\phi}$$

Here ϕ is **not** a dispersion parameter but a **precision parameter**!

Beta regression is suitable when the outcome is restricted to interval (a, b)
that can be transferred to unit interval with transformation $\frac{y-a}{b-a}$

Noise in Beta regression model is heteroscedastic

Beta regression usually uses logit link

Example: Beta-regression

Let's simulate data from Beta regression model

$$\mathbf{y}_i \sim \text{Beta}(\mu_i, \phi) = f(y_i \mid \mu_i, \phi) = \frac{\Gamma(\phi)}{\Gamma(\mu_i\phi)\Gamma((1-\mu_i)\phi)} y_i^{\mu_i\phi-1} (1-y_i)^{(1-\mu_i)\phi-1}$$

$$\eta_i = x_i^T \beta$$

$$\mu_i = g^{-1}(\eta_i) = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

$$\mathbb{E}(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \frac{\mu_i(1-\mu_i)}{1+\phi}$$

Example: Beta-regression

Let's simulate data from Beta regression model

$$\mathbf{y}_i \sim \text{Beta}(\mu_i, \phi) = f(y_i \mid \mu_i, \phi) = \frac{\Gamma(\phi)}{\Gamma(\mu_i\phi)\Gamma((1-\mu_i)\phi)} y_i^{\mu_i\phi-1} (1-y_i)^{(1-\mu_i)\phi-1}$$

$$\eta_i = x_i^T \beta$$

$$\mu_i = g^{-1}(\eta_i) = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

$$E(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \frac{\mu_i(1-\mu_i)}{1+\phi}$$

Compare the common beta distribution formulation (also used in R) and the mean & precision - formulation above

Common:

$$f(y_i \mid \alpha_i, \beta_i) = \frac{\Gamma(\alpha_i + \beta_i)}{\Gamma(\alpha_i)\Gamma(\beta_i)} y_i^{\alpha_i-1} (1-y_i)^{\beta_i-1}$$

where

$$\alpha_i = \mu_i\phi$$

$$\beta_i = (1-\mu_i)\phi$$

Maximum likelihood estimation

Exponential family of distributions (EFD)

The distribution of a random variable y_i belongs to the exponential family if its probability density function can be written in the form

$$f(y_i | \theta_i, \phi) = \exp \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right)$$

θ_i the natural, or canonical, parameter

ϕ the dispersion parameter

$b(\theta_i)$ function of the natural parameter

$a(\phi)$ function of the dispersion parameter

$c(y_i, \phi)$ function of the data value and the dispersion

Exponential family of distributions (EFD)

With GLMs we use this following form for the exponential family of distributions

$$f(y_i | \theta_i, \phi) = \exp \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right)$$

that includes the dispersion parameter ϕ explicitly.

In many sources you also often see the formulation

$$f(y_i | \theta_i) = h(y_i) \exp(\eta(\theta_i)T(y_i) - A(\theta_i))$$

that is the basic exponential family representation

Exponential family of distributions (EFD)

If the distribution is a **single-parameter** distribution such as **Bernoulli** and **Poisson** distribution, then there is no dispersion parameter and the format is

$$f(y_i | \theta_i, \phi) = \exp(y_i \theta_i - b(\theta_i) + c(y_i))$$

θ_i the natural, or canonical, parameter

~~ϕ the dispersion parameter~~

$b(\theta_i)$ function of the natural parameter

~~$a(\phi)$ function of the dispersion parameter~~

~~$c(y_i, \phi)$ function of the data value and the dispersion~~

Exponential family of distributions (EFD)

Likelihood $L(\theta_i, \phi) = \exp \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right)$

Log-likelihood $l(\theta_i, \phi) = \log(L(\theta_i, \phi)) = \frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)$

The mean and the variance of these distributions are of the form

$$\mu_i = E(\mathbf{y}_i \mid x_i) = b'(\theta_i), \text{Var}(\mathbf{y}_i \mid x_i) = b''(\theta_i)a(\phi)$$

Canonical link function is defined by $g(\mu_i) = \theta_i$

Using canonical link means $\eta_i = x_i^T \beta = \theta_i, \quad \mu_i = b'(\eta_i)$

Mean of a distribution from EFD

Calculate first the partial derivative $\frac{\partial l(\theta_i, \phi)}{\partial \theta_i} = \frac{y_i - \frac{\partial b(\theta_i)}{\partial \theta_i}}{a(\phi)}$

Calculate a property of log-likelihoods
$$\begin{aligned} \mathbb{E} \left(\frac{\partial l(\theta_i, \phi)}{\partial \theta_i} \right) &= \mathbb{E} \left(\frac{\partial}{\partial \theta_i} \log f(y_i | \theta_i, \phi) \right) = \mathbb{E} \left(\frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial}{\partial \theta_i} f(y_i | \theta_i, \phi) \right) \\ &= \int f(y_i | \theta_i, \phi) \cdot \frac{1}{f(y_i | \theta_i, \phi)} \cdot \frac{\partial}{\partial \theta_i} f(y_i | \theta_i, \phi) dy_i \\ &= \frac{\partial}{\partial \theta_i} \int f(y_i | \theta_i, \phi) dy_i = 0 \end{aligned}$$

Combine the results
$$\begin{aligned} 0 &= \mathbb{E} \left(\frac{\partial l(\theta_i, \phi)}{\partial \theta_i} \right) = \mathbb{E} \left(\frac{\mathbf{y}_i - \frac{\partial b(\theta_i)}{\partial \theta_i}}{a(\phi)} \right) = \frac{\mathbb{E}(\mathbf{y}_i) - \frac{\partial b(\theta_i)}{\partial \theta_i}}{a(\phi)} \\ &\Leftrightarrow \mathbb{E}(\mathbf{y}_i) = \frac{\partial b(\theta_i)}{\partial \theta_i} = b'(\theta_i) \end{aligned}$$

Variance of a distribution from EFD

$$\begin{aligned} \mathbb{E} \left(\frac{\partial^2 l(\theta_i, \phi)}{\partial \theta_i^2} \right) &= \mathbb{E} \left(\frac{\partial}{\partial \theta_i} \left(\frac{\partial}{\partial \theta_i} \log f(y_i | \theta_i, \phi) \right) \right) \\ &= \mathbb{E} \left(\frac{\partial}{\partial \theta_i} \left(\frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial}{\partial \theta_i} f(y_i | \theta_i, \phi) \right) \right) \\ &= \mathbb{E} \left(-\frac{1}{f(y_i | \theta_i, \phi)^2} \left(\frac{\partial}{\partial \theta_i} f(y_i | \theta_i, \phi) \right)^2 + \frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial^2}{\partial \theta_i^2} f(y_i | \theta_i, \phi) \right) \\ &= -\mathbb{E} \left(\left(\frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial}{\partial \theta_i} f(y_i | \theta_i, \phi) \right)^2 \right) + \mathbb{E} \left(\frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial^2}{\partial \theta_i^2} f(y_i | \theta_i, \phi) \right) \\ &\quad \swarrow \\ &= \mathbb{E} \left(\frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial^2}{\partial \theta_i^2} f(y_i | \theta_i, \phi) \right) = \int f(y_i | \theta_i, \phi) \frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial^2}{\partial \theta_i^2} f(y_i | \theta_i, \phi) \mathrm{d}y_i \\ &= \frac{\partial^2}{\partial \theta_i^2} \int f(y_i | \theta_i, \phi) \mathrm{d}y_i = \frac{\partial^2}{\partial \theta_i^2} 1 \mathrm{d}y_i = 0 \end{aligned}$$

Variance of a distribution from EFD

$$\begin{aligned}
 \mathbb{E} \left(\frac{\partial^2 l(\theta_i, \phi)}{\partial \theta_i^2} \right) &= \mathbb{E} \left(\frac{\partial}{\partial \theta_i} \left(\frac{\partial}{\partial \theta_i} \log f(y_i | \theta_i, \phi) \right) \right) \\
 &= \mathbb{E} \left(\frac{\partial}{\partial \theta_i} \left(\frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial}{\partial \theta_i} f(y_i | \theta_i, \phi) \right) \right) \\
 &= \mathbb{E} \left(-\frac{1}{f(y_i | \theta_i, \phi)^2} \left(\frac{\partial}{\partial \theta_i} f(y_i | \theta_i, \phi) \right)^2 + \frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial^2}{\partial \theta_i^2} f(y_i | \theta_i, \phi) \right) \\
 &= -\mathbb{E} \left(\left(\frac{1}{f(y_i | \theta_i, \phi)} \frac{\partial}{\partial \theta_i} f(y_i | \theta_i, \phi) \right)^2 \right) \\
 &= -\mathbb{E} \left(\left(\frac{\partial}{\partial \theta_i} l(\theta_i, \phi) \right)^2 \right)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial l(\theta_i, \phi)}{\partial \theta_i} &= \frac{y_i - \frac{\partial b(\theta_i)}{\partial \theta_i}}{a(\phi)} \\
 \frac{\partial^2 l(\theta_i, \phi)}{\partial \theta_i^2} &= -\frac{1}{a(\phi)} \frac{\partial^2 b(\theta_i)}{\partial \theta_i^2}
 \end{aligned}$$

$$\begin{aligned}
 -\mathbb{E} \left(\frac{1}{a(\phi)} \frac{\partial^2 b(\theta_i)}{\partial \theta_i^2} \right) &= -\mathbb{E} \left(\left(\frac{\mathbf{y}_i - \frac{\partial b(\theta_i)}{\partial \theta_i}}{a(\phi)} \right)^2 \right) \\
 \Leftrightarrow \mathbb{E} \left(\left(\mathbf{y}_i - \frac{\partial b(\theta_i)}{\partial \theta_i} \right)^2 \right) &= \text{Var}(\mathbf{y}_i) = a(\phi) \frac{\partial^2 b(\theta_i)}{\partial \theta_i^2}
 \end{aligned}$$

End of Lecture 4